

Assignment 1

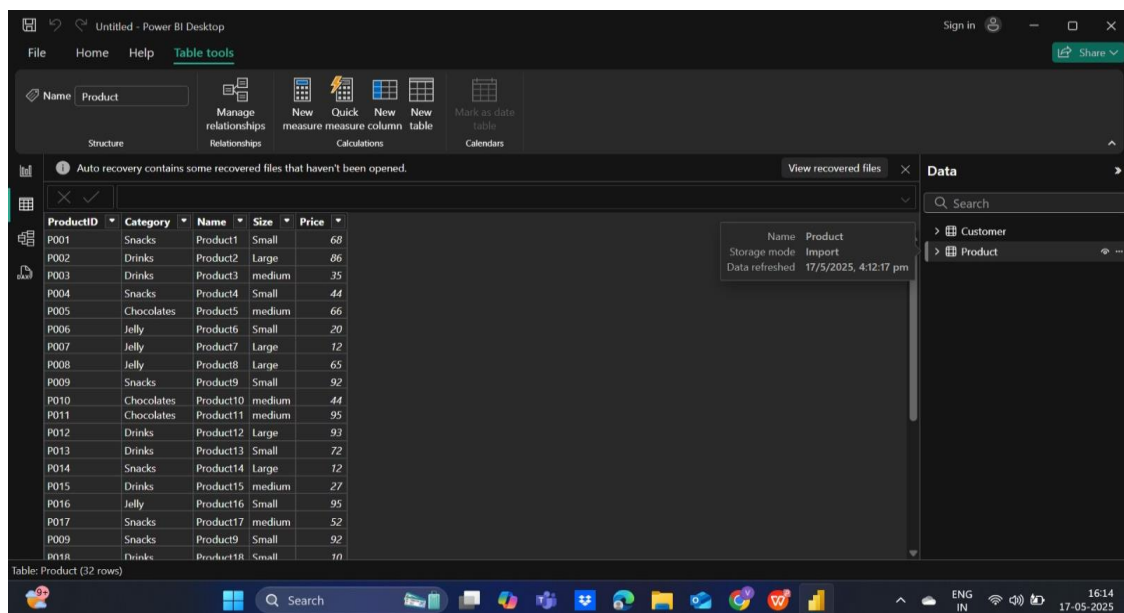
Task 1:

- Round the 'Price' column in the Product dataset to the nearest integer for simplicity.
- Split the 'Customer' column in the Customer table into two columns: 'FirstName' and 'LastName'.

Sol:

Step1:

- The Price column in the Product dataset originally contained floating-point (decimal) numbers.
- Rounding the prices makes the data cleaner and easier to interpret, especially in visual dashboards or summary tables.
- Using a rounding function (in Power BI), we converted all decimal price values to their nearest integer.



The screenshot shows the Power BI Desktop interface. The 'Table tools' ribbon is active, and the 'Product' table is selected. The table view displays the following data:

ProductID	Category	Name	Size	Price
P001	Snacks	Product1	Small	68
P002	Drinks	Product2	Large	86
P003	Drinks	Product3	medium	35
P004	Snacks	Product4	Small	44
P005	Chocolates	Product5	medium	66
P006	Jelly	Product6	Small	20
P007	Jelly	Product7	Large	12
P008	Jelly	Product8	Large	65
P009	Snacks	Product9	Small	92
P010	Chocolates	Product10	medium	44
P011	Chocolates	Product11	medium	95
P012	Drinks	Product12	Large	93
P013	Drinks	Product13	Small	72
P014	Snacks	Product14	Large	12
P015	Drinks	Product15	medium	27
P016	Jelly	Product16	Small	95
P017	Snacks	Product17	medium	52
P018	Snacks	Product18	Small	92
P019	Drinks	Product19	Small	10

The 'Data' pane on the right shows the 'Product' table selected. The status bar at the bottom indicates 'Table: Product (32 rows)'.

Step2:

- The original Customer column stored names as a single string (e.g., "Sujata Mohanty").
- For better personalization, sorting, filtering, and analysis, we split this into two separate fields: FirstName and LastName.

- This was achieved by splitting the text using a space character as the delimiter, assuming the first word is the first name and the second is the last name.

CustomerID	FirstName	LastName	Gender	Area	Profession
C0001	Sujata	Mohanty	Male	middle	Retired
C0002	Suraj	Rajput	Male	east	unemployment
C0003	Pramod	Bhavsar	Male	east	profession
C0004	Satish	Ojha	Male	west	self-employed
C0005	Suntu	Kumar	Male	middle	Retired
C0006	Krutika	Shelar	Male	middle	unemployment
C0007	Arjun	Shaw	Male	east	profession
C0008	Suraj	Rajput	Male	east	unemployment
C0009	Shrikant	Badge	Female	west	self-employed
C0010	Jitender	Kumar	Male	south	Retired
C0011	Dharmendar	Rana	Male	middle	unemployment
C0012	Adnan	Soukat	Female	south	profession
C0013	Sheetal	Nishad	Male	middle	self-employed
C0014	Monika	Pawar	Female	east	Retired
C0015	Meena	Mourya	Male	east	unemployment
C0016	Ashu	Sharma	Male	west	profession
C0017	Hariyansh	Gautam	Male	middle	self-employed
C0018	Vini	Saini	Female	middle	Retired
C0019	Anand	Singh Rainut	Male	east	unemployment

Task 2:

- Convert all entries in the 'Category' column in the Product table to uppercase.
- Replace all occurrences of 'unemployment' with 'Unemployed' in the 'Profession' column of the Customer table.

Sol:

Step 1:

- Changed all entries in the 'Category' column (e.g., "Drinks", "Snacks") to uppercase (e.g., "DRINKS", "SNACKS").
- This improves consistency and avoids mismatches caused by different text casing during grouping or filtering.

Table: Product (32 rows)

ProductID	Category	Name	Size	Price
P001	SNACKS	Product1	Small	68
P002	DRINKS	Product2	Large	86
P003	DRINKS	Product3	medium	35
P004	SNACKS	Product4	Small	44
P005	CHOCOLATES	Product5	medium	66
P006	JELLY	Product6	Small	20
P007	JELLY	Product7	Large	12
P008	JELLY	Product8	Large	65
P009	SNACKS	Product9	Small	92
P010	CHOCOLATES	Product10	medium	44
P011	CHOCOLATES	Product11	medium	95
P012	DRINKS	Product12	Large	93
P013	DRINKS	Product13	Small	72
P014	SNACKS	Product14	Large	12
P015	DRINKS	Product15	medium	27
P016	JELLY	Product16	Small	95
P017	SNACKS	Product17	medium	52
P018	SNACKS	Product18	Small	92
P019	DRINKS	Product19	Small	10

Step 2:

- In the 'profession' column, replaced all instances of "unemployment" with "Unemployed".
- To ensure data clarity and use standard terminology for better readability and reporting.

Table: Customer (32 rows)

CustomerID	FirstName	LastName	Gender	Area	Profession
C0001	Sujata	Mohanty	Male	middle	Retired
C0002	Suraj	Rajput	Male	east	Unemployed
C0003	Pramod	Bhavsar	Male	east	profession
C0004	Satish	Ojha	Male	west	self-employed
C0005	Sintu	Kumar	Male	middle	Retired
C0006	Krutika	Shelar	Male	middle	Unemployed
C0007	Arjun	Shaw	Male	east	profession
C0008	Shrikant	Badge	Female	west	self-employed
C0009	Jitender	Kumar	Male	south	Retired
C0010	Dharmendar	Rana	Male	middle	Unemployed
C0011	Adnan	Soukat	Female	south	profession
C0012	Sheetal	Nishad	Male	middle	self-employed
C0013	Monika	Pawar	Female	east	Retired
C0014	Meena	Mourya	Male	east	Unemployed
C0015	Ashu	Sharma	Male	west	profession
C0016	Harivansh	Gautam	Male	middle	self-employed
C0017	Vini	Saini	Female	middle	Retired
C0018	Anand	Singh Rajput	Male	east	Unemployed
C0019	Jaishree	Kumar	Female	west	profession

Task 3:

- Ensure all columns in the datasets have appropriate data types, e.g., 'Date' column as Date type, 'Price' as Decimal type.
- Identify and replace any inconsistent values in the 'Size' column of the Product dataset to ensure uniformity (e.g., replace "medium" with "Medium").

Sol:

Verified the data types of each column in the Product and Customer datasets.

Ensured the following:

Columns like Price are stored as Decimal (or Numeric) types.

Columns representing dates (e.g., Date) are set as Date or Datetime types.

Replaced “medium” with “Medium”.

ProductID	Category	Name	Size	Price
1	SNACKS	Product1	Small	66
2	DRINKS	Product2	Large	86
3	DRINKS	Product3	Medium	35
4	SNACKS	Product4	Small	44
5	CHOCOLATES	Product5	Medium	66
6	JELLY	Product6	Small	20
7	JELLY	Product7	Large	12
8	JELLY	Product8	Large	65
9	SNACKS	Product9	Small	92
10	CHOCOLATES	Product10	Medium	44
11	CHOCOLATES	Product11	Medium	95
12	DRINKS	Product12	Large	93
13	DRINKS	Product13	Small	22
14	SNACKS	Product14	Large	12
15	DRINKS	Product15	Medium	27
16	JELLY	Product16	Small	95

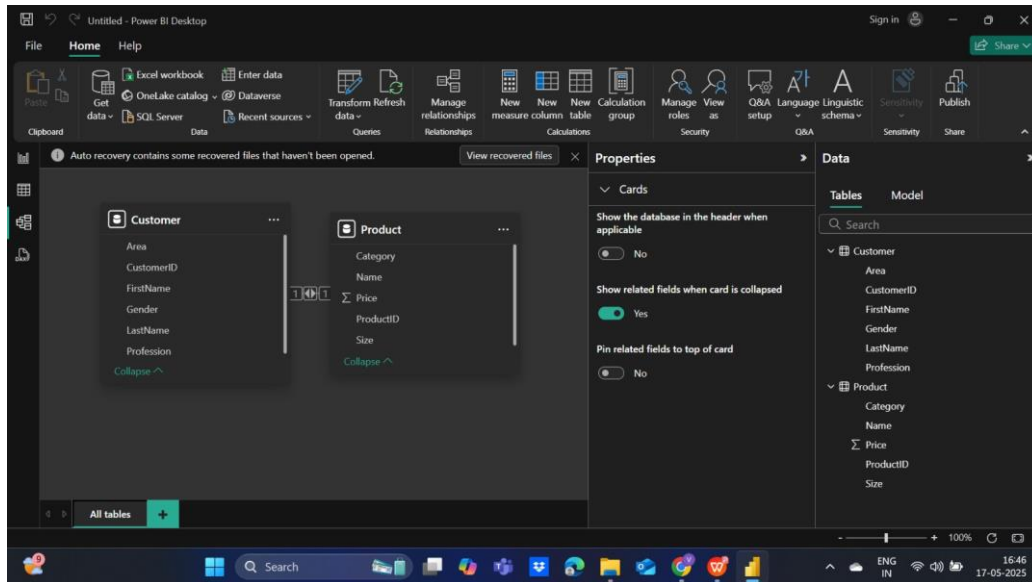
Task 4:

- Create relationships between the tables using 'CustomerID' and 'ProductID' as keys.
- Clean the data by removing any duplicate entries in the Customer and Product tables.

Sol:

- Imported all three tables into Power BI: Customer, Product, and Sales.
- Used the Model View in Power BI to establish the following relationships:
- Customer[CustomerID] → Sales[CustomerID]
- Product[ProductID] → Sales[ProductID]
- Set the relationships as One-to-One:
- One Customer can appear in one Sales records.
- One Product can be sold in one Sales records.
- Ensured relationships are active and cross-filtering is set to single direction for performance efficiency.

- Established a relational model that allows filtering, aggregation, and joining of fields across datasets.
- Enables advanced visualizations like sales by customer demographics or product category.
- All entries in the Customer and Product tables are now unique.



Task 5:

Sales by Category:

- Create a chart showing total sales (TotalPrice) by product category.
- Customize colors, refine the title, and add data labels for exact sales amounts.

Sales Percentage by Category:

- Create a chart showing the percentage share of sales (Price) for each product category.
- Keep the chart compact and easy to understand.

Sol:

This Power BI dashboard integrates cleaned, structured data with effective visualizations to provide a comprehensive view of customer and product performance. The data model and visuals support strategic decision-making by highlighting trends, key categories, and revenue insights.

